

"What is the overall correlation between vaccination rates and reduction in COVID-19 cases and deaths across top-affected countries? (2020-2023)"

By : Purnima Adhikari

Introduction

- COVID-19 outcomes varied across countries due to differences in vaccination, healthcare, and public health policies.
- This project analyzes global trends to understand how vaccination levels impact death rates.
- Goal: Identify the strongest factors influencing COVID-19 mortality globally.

Materials

- JHU CSSE COVID-19 Data (GitHub)
- GovEx Global Vaccine Data (GitHub)
- GHS Index data from ghsindex.org
- Google Collab
- Python (libraries included)

Methodology

- Merged and cleaned datasets.

Normalized by population:

- Cases/deaths per million
- Vaccine doses per 100 people

- Selected 5–10 key countries for focus.

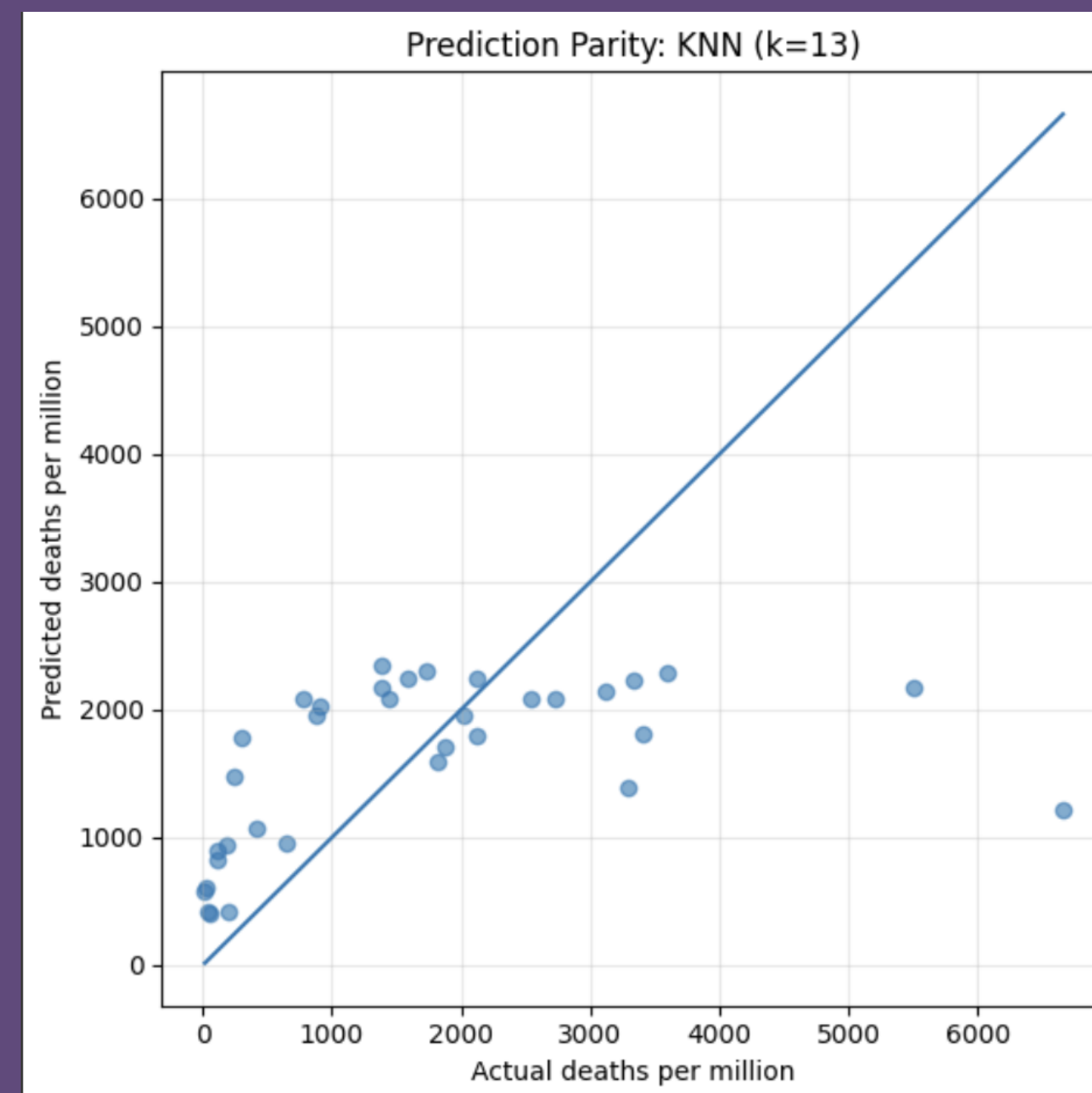
- Conducted correlation and machine learning modeling (Linear Regression, KNN).

- Modeled outbreak dynamics with an SIR model.

Results

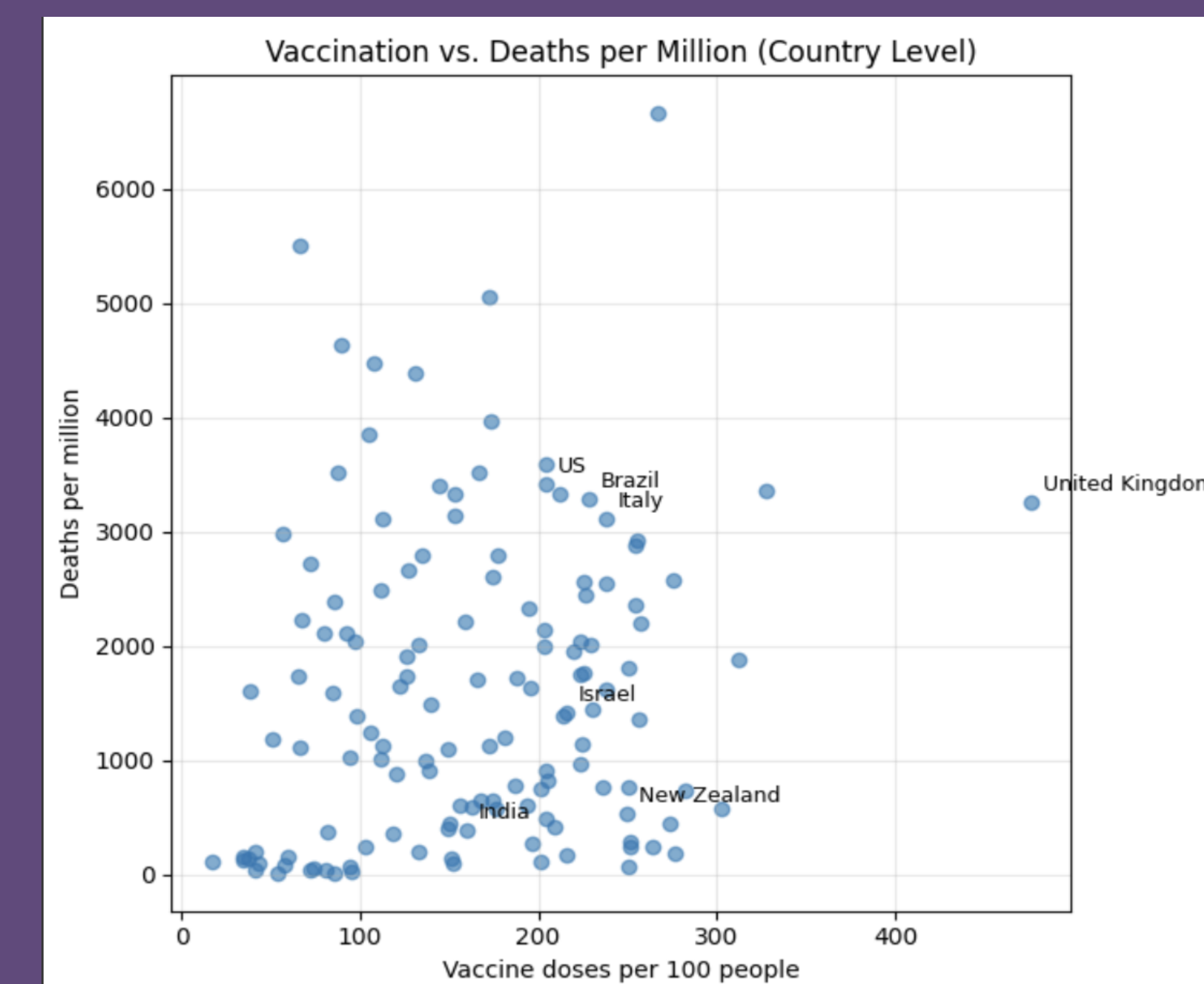
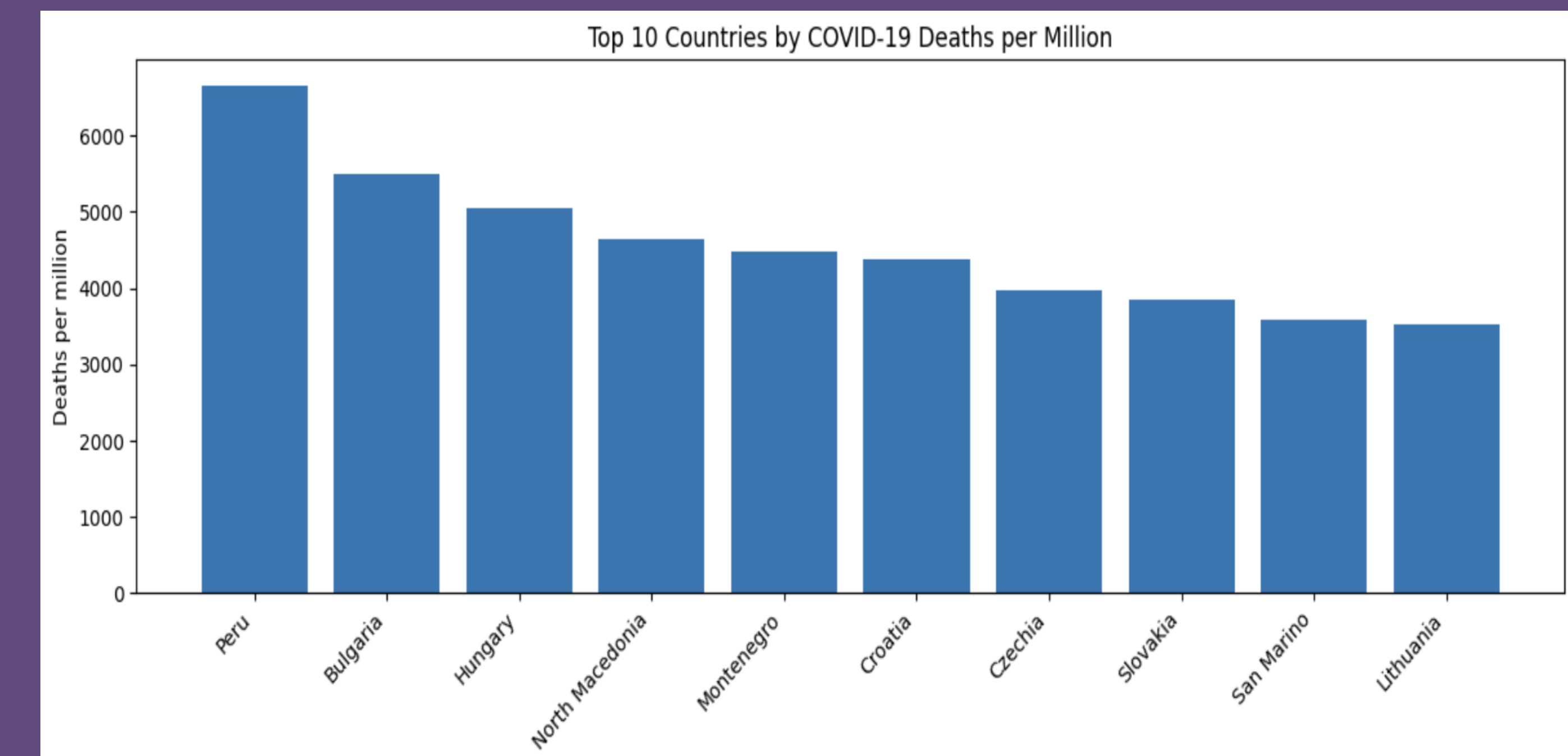
Machine Learning Models

- Linear Regression $R^2 \approx 0.63$
- KNN $R^2 \approx 0.67$ (better fit for non-linear trends)
- Most predictive feature: vaccine doses per 100 people



Top 10 Countries by Deaths per Million

- Countries like Peru, Hungary, and Czechia had highest normalized deaths
- Not always linked to low vaccination, revealing other contributing factors

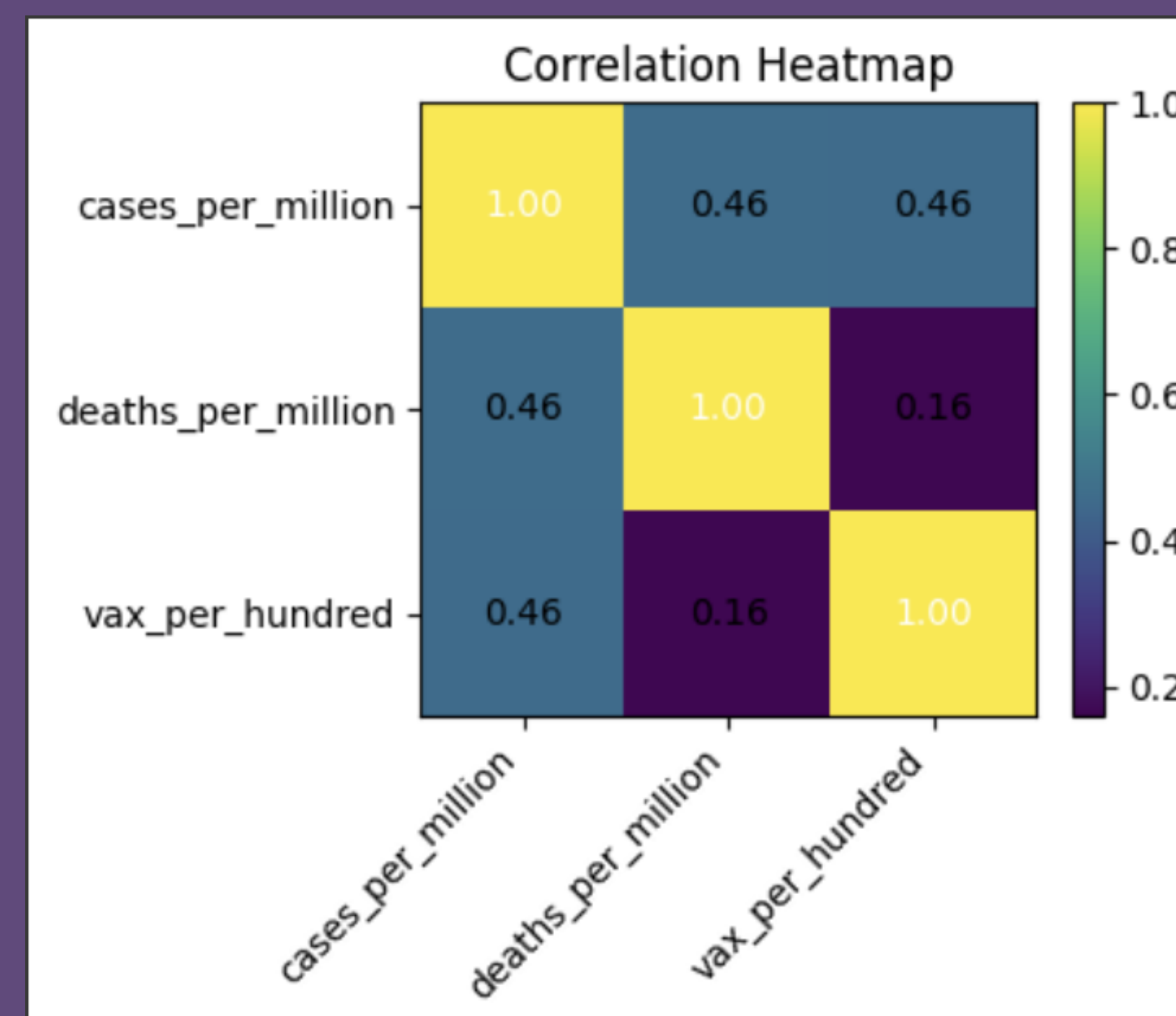


Vaccine vs Death Correlation

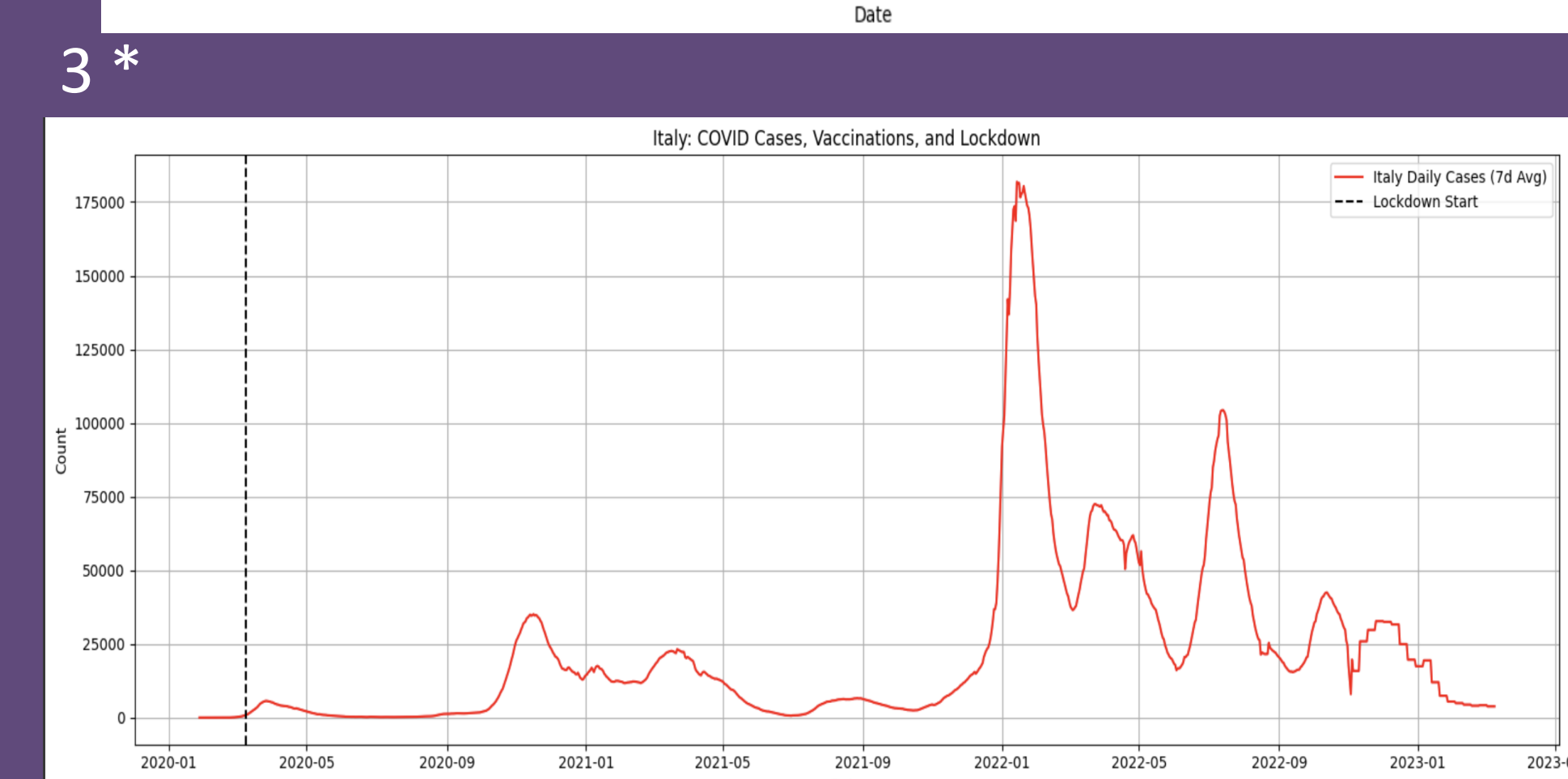
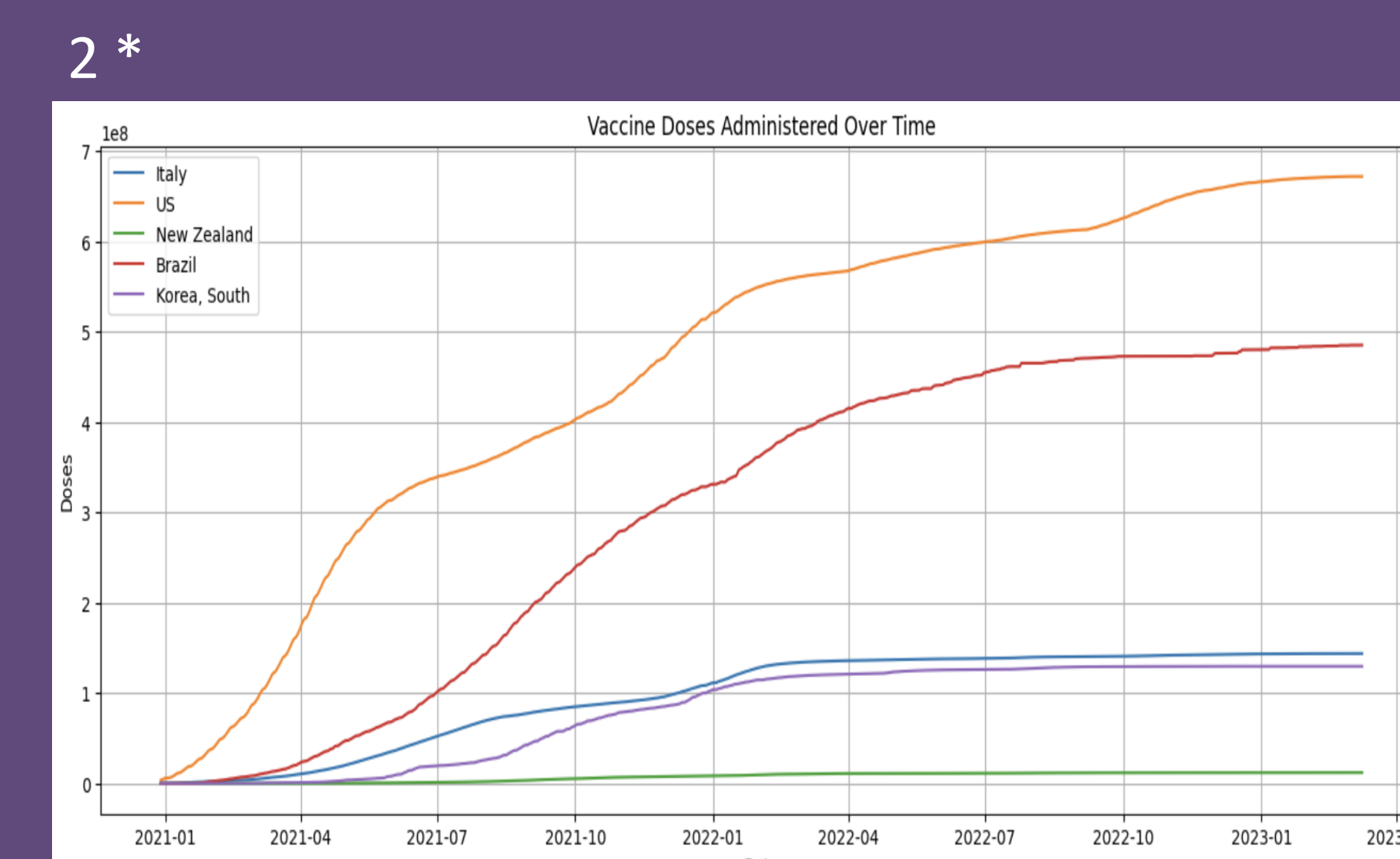
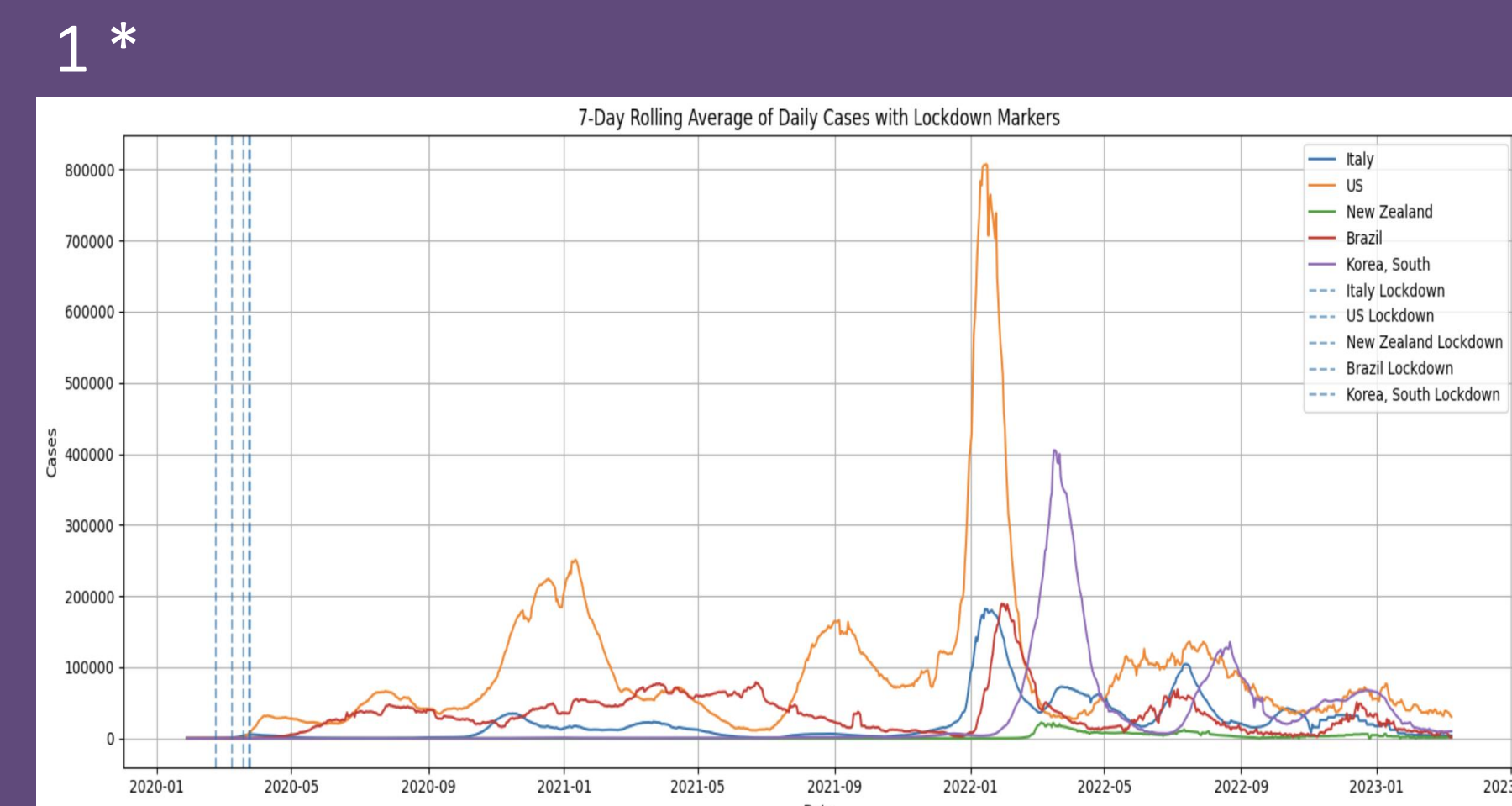
- Strong negative correlation : higher vaccination → lower death rates
- UK and Israel show sharp mortality drops post-vaccine
- Brazil shows a weaker correlation due to delayed rollouts

Correlation Matrix

- Vaccination rate has the highest negative correlation with deaths
- Cases and deaths positively correlated, as expected
- Preparedness index did not correlate strongly with outcomes



EXTRA GRAPHS :



Conclusion

- Vaccination rate is the strongest predictor of reduced COVID-19 deaths.
- Countries with earlier and broader rollouts saw better outcomes.
- Preparedness scores didn't consistently match real-world results.
- Possibility : (SIR model effectively simulates how public health actions alter transmission dynamics.)

Future Work

Next Steps:

- Extend to newer variants (e.g., Omicron)
- Explore socioeconomic impacts
- Cluster countries using unsupervised learning
- SIR Model for most impacted countries

References

- Dong E, Du H, Gardner L. An interactive web-based dashboard to track COVID-19 in real time. *Lancet Inf Dis*. 2020(5):533-534. doi: 10.1016/S1473-3099(20)30120-1
- John Hopkins University
- JHU CSSE, GovEx, GHS Index
- The Coding School
- TCS TA : Durrya
- Project Mentor : Padmavathy K. Kumar